



Assignment: 01

Course Title: Project on Web Development

Course Code: SWE-382

Experiment Name: MNEXPharma - Pharmacy Management System

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MNEXPharma - Pharmacy Management System

Abstract

MNEXPharma is a modern web-based Pharmacy Management System developed to streamline pharmacy operations through a centralized digital platform. The system provides an intuitive interface for managing medicines, inventory, suppliers, customers, prescriptions, sales transactions, employees, and business reports.

The project focuses on creating a responsive and user-friendly dashboard that allows pharmacy staff to efficiently monitor stock levels, process customer transactions, manage supplier information, and analyze business performance. The system has been developed using Vue.js, Tailwind CSS, Pinia, and Vue Router, providing a scalable foundation for future backend integration using Laravel and MySQL.

1. Introduction

Pharmacies handle large volumes of medicines, prescriptions, inventory records, supplier transactions, and customer information on a daily basis. Traditional manual record-keeping systems are often inefficient, time-consuming, and prone to human errors.

MNEXPharma was developed to provide a modern digital solution for pharmacy management. The system centralizes operational activities into a single platform, enabling better organization, faster access to information, and improved decision-making.

The application is designed with a responsive interface that ensures usability across various devices while maintaining a professional and intuitive user experience.

2. Project Objectives

The primary objectives of this project are:

- To digitize pharmacy management operations.
- To improve medicine inventory tracking.

- To simplify sales and billing processes.
 - To manage suppliers and customers efficiently.
 - To monitor medicine expiry dates and stock levels.
 - To provide business analytics and reporting.
 - To create a scalable platform for future backend development.
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3. Project Overview

MNEXPharma is a complete frontend prototype of a Pharmacy Management System designed for pharmacies, medical stores, and healthcare organizations.

The system consists of multiple interconnected modules that support various business operations including medicine management, inventory tracking, supplier management, customer records, prescription processing, and sales transactions.

The platform uses a modern dashboard interface that provides real-time operational insights through visual analytics, summary cards, reports, and activity tracking.

4. System Modules

The application contains the following modules:

- Dashboard
 - Medicines Management
 - Inventory Management
 - Sales POS (Point of Sale)
 - Purchase Management
 - Supplier Management
 - Customer Management
 - Prescription Management
 - Doctor Management
 - Reports & Analytics
 - Employee Management
 - Notifications
 - Users & Roles
 - Settings
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5. Feature List

Dashboard

The dashboard serves as the central control panel of the system.

Features:

- Revenue overview
 - Sales analytics
 - Inventory statistics
 - Low stock monitoring
 - Expired medicine tracking
 - Recent sales overview
 - Category distribution analysis
 - Business performance insights
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Medicines Management

The Medicines module allows management of pharmacy products and stock.

Features:

- Medicine catalog
 - Search and filtering system
 - Medicine categorization
 - Supplier information tracking
 - Stock monitoring
 - Expiry date tracking
 - Status management
 - Medicine details management
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Inventory Management

The Inventory module helps maintain accurate stock levels.

Features:

- Inventory monitoring
 - Stock level tracking
 - Reorder alerts
 - Inventory value tracking
 - Expiry monitoring
 - Inventory activity timeline
 - Stock adjustment management
 - Low stock notifications
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Sales POS (Point of Sale)

The POS module manages customer purchases and billing.

Features:

- Medicine selection system
 - Shopping cart functionality
 - Customer billing
 - Discount calculation
 - VAT calculation
 - Multiple payment methods
 - Invoice generation
 - Sales history tracking
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Purchase Management

The Purchase module manages supplier purchases.

Features:

- Purchase tracking
 - Supplier purchase records
 - Payment monitoring
 - Purchase status management
 - Order history
 - Delivery tracking
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Supplier Management

The Supplier module manages supplier information.

Features:

- Supplier database
 - Contact management
 - Purchase history
 - Outstanding payment tracking
 - Supplier status management
 - Business relationship records
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Customer Management

The Customer module maintains customer information.

Features:

- Customer records
- Purchase history
- Customer analytics

- Revenue contribution tracking
 - Customer categorization
 - Contact information management
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Prescription Management

The Prescription module manages patient prescriptions.

Features:

- Prescription records
 - Verification workflow
 - Dispensing management
 - Prescription status tracking
 - Patient information management
 - Search and filtering
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Doctor Management

The Doctor module stores physician information.

Features:

- Doctor profiles
 - Specialization records
 - Contact information
 - Medical reference management
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Reports & Analytics

The Reports module provides business insights.

Features:

- Revenue reports
 - Sales reports
 - Inventory reports
 - Category analysis
 - Monthly performance tracking
 - Business analytics dashboards
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Employee Management

The Employee module manages staff information.

Features:

- Employee records
 - Staff management
 - Role assignment
 - Status tracking
 - Employee information storage
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Notifications

The Notification system provides operational alerts.

Features:

- Low stock alerts
- Expiry notifications
- Purchase updates
- System activity notifications

Settings

The Settings module controls application configurations.

Features:

- Pharmacy information management
 - System preferences
 - Application configuration
 - User settings
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6. Technology Stack

The project was developed using modern frontend technologies.

Frontend Technologies:

- Vue.js 3
- Vite
- Tailwind CSS
- Pinia
- Vue Router
- Chart.js

Development Tools:

- Visual Studio Code
- Claude Code
- npm Package Manager

Future Backend Technologies:

- PHP Laravel
- MySQL Database

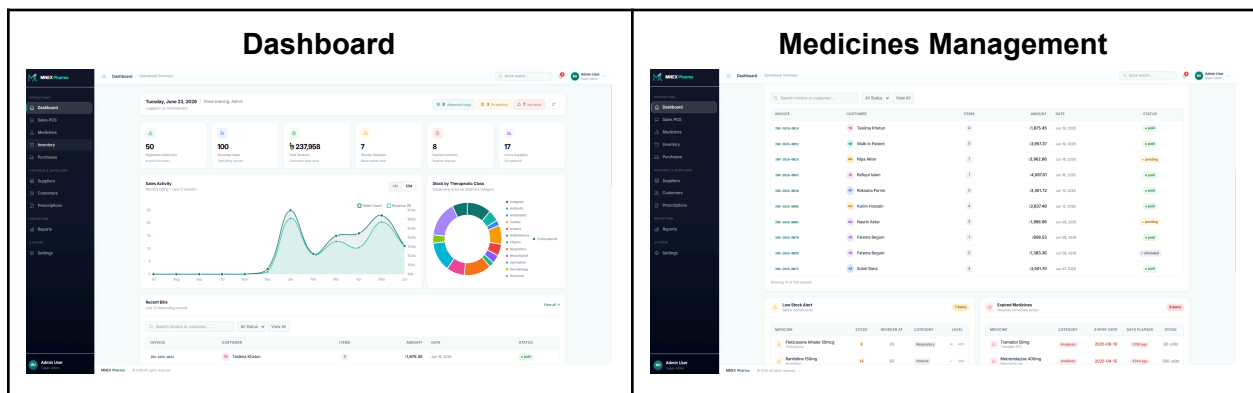
7. User Interface Design

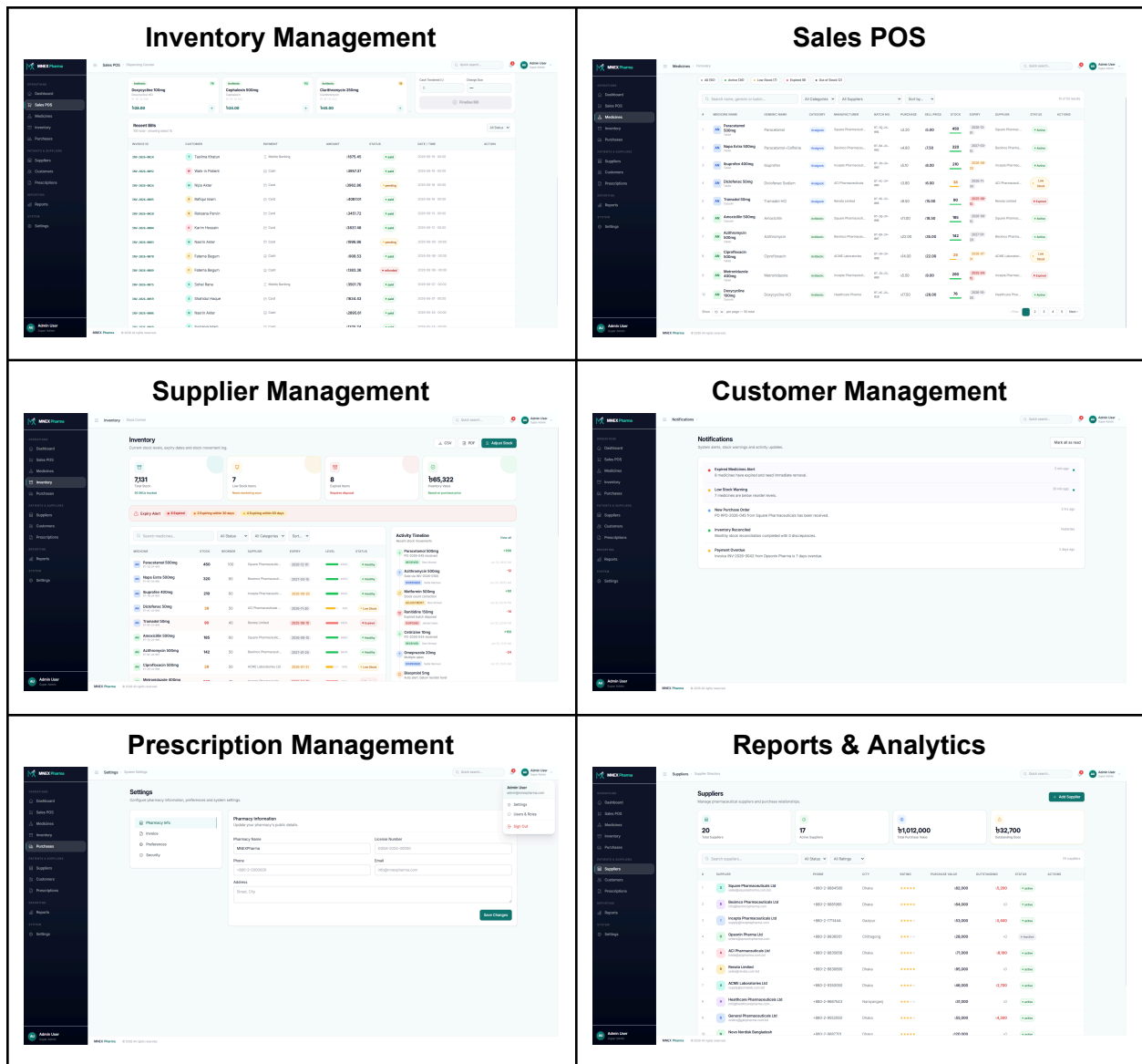
The user interface was designed using modern dashboard principles to ensure usability, accessibility, and responsiveness.

Key Design Features:

- Responsive layout
- Sidebar navigation
- Dashboard analytics
- Interactive tables
- Modern card-based design
- Visual data representation
- Mobile-friendly structure
- Professional healthcare-themed interface

8. Screenshots





9. Future Scope

The project can be enhanced with the following future developments:

- Laravel backend integration
- MySQL database implementation
- User authentication system

- Barcode scanner support
 - Multi-branch pharmacy management
 - Cloud deployment
 - Real-time inventory synchronization
 - Automated report generation
 - Online prescription processing
 - Mobile application development
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10. Conclusion

MNEXPharma successfully demonstrates a complete Pharmacy Management System user interface capable of supporting major pharmacy operations. The project provides a scalable and well-structured foundation for future backend integration and real-world implementation.

Through its modern dashboard, responsive design, and modular architecture, the system offers an efficient solution for managing pharmacy activities, improving operational efficiency, and enhancing user experience. The project showcases practical application of modern web technologies in solving real-world business management challenges.